

Marketing Campaign ROI Dashboard

A/B Test Analysis Across 5 Channels — Insights, Trends & Budget Recommendations

Facebook · Google Search · Instagram · Email · TikTok

Tech stack: R (data cleaning & statistics) + Tableau (visualization)

Executive Summary

Over a 90-day period, the business spent \$35,749.84 across five paid channels, generating \$658,172.02 in attributed revenue — a blended ROAS of 18.4x and a blended CAC of \$2.95 per conversion. Performance, however, is highly uneven across channels: Email dwarfs every paid channel on efficiency, while three of the four paid-media channels (Facebook, Instagram, TikTok) show a statistically significant winning creative (Variant B), and Google Search shows no significant difference between variants.

\$35,749.84 Total Spend	\$658,172.02 Total Revenue	18.4x Blended ROAS	\$2.95 Blended CAC
-----------------------------------	--------------------------------------	------------------------------	------------------------------

Three headline takeaways:

- Email is the single most efficient channel by a wide margin (ROAS 510.8x, CAC \$0.11) — but the original creative (Variant A) significantly outperforms the new one (Variant B) here, the opposite pattern of every other channel.
- TikTok, Facebook, and Instagram all show Variant B (the new creative) winning with statistical significance — the new creative genuinely outperforms the original on paid social.
- TikTok is the fastest-growing channel over the test window (+56% weekly impression volume from week 1 to week 13) while Google Search shows creative fatigue (click-through rate down ~12% over the same window).

Methodology

The underlying pipeline (built in R) simulates a realistic ad-platform export, then cleans and analyzes it:

- Standardized inconsistent channel naming (e.g. "FB", "facebook", "Facebook" → "Facebook").
- Parsed mixed date formats using a fixed priority order (ISO, then MM/DD/YYYY, then DD-MM-YYYY) to avoid guessing on ambiguous dates.
- Removed exact duplicate rows and rows where conversions exceeded clicks (tracking glitches).
- Imputed missing spend values using the channel + variant median rather than dropping rows, to avoid biasing CAC by removing real conversions from the denominator context.
- Computed CTR, conversion rate, CAC (spend ÷ conversions), and ROAS (revenue ÷ spend) at the daily grain, then rolled up to channel and channel × variant summaries.
- Tested Variant A vs. B conversion-rate differences per channel with a two-proportion z-test (95% confidence).

Data quality note: roughly 2% of rows carry dates outside the intended Jan 1 – Mar 30, 2024 window, straying as far as October–December. This traces back to genuine ambiguity between DD-MM-YYYY and MM-DD-YYYY when both day and month are 12 or below — no fixed parsing order can resolve that case with certainty from the value alone. These rows are immaterial to the totals but are a useful reminder that date format should be locked at the source system, not reconstructed after export.

Channel-Level Performance

Ranked by CAC (lowest, i.e. most efficient, first):

Channel	Spend	Conversions	Revenue	CAC	ROAS	Conv. Rate
Email	\$535.14	5,068	\$273,367.20	\$0.11	510.8x	11.41%
Google Search	\$10,806.66	3,147	\$170,850.14	\$3.43	15.8x	8.71%
TikTok	\$6,551.54	1,411	\$76,972.22	\$4.64	11.75x	3.43%
Facebook	\$9,034.55	1,463	\$80,993.60	\$6.18	8.96x	4.87%
Instagram	\$8,821.95	1,020	\$55,988.86	\$8.65	6.35x	3.52%

Email converts at a far higher rate and near-zero cost per click (it's a house list, not an auction), which mechanically produces an outsized ROAS relative to paid social/search — it should be read as "extremely efficient owned channel," not as a template for what paid acquisition costs should look like. Among the paid-media channels, Google Search is currently the most efficient on a pure CAC basis, with TikTok close behind and improving quickly (see Trends). Instagram is the weakest channel on every efficiency metric.

A/B Test Results by Channel

Two-proportion z-test on conversion rate, Variant A vs. Variant B, 95% confidence:

Channel	Variant A CVR	Variant B CVR	p-value	Significant?	Winner
Facebook	4.50%	5.15%	0.0114	Yes	B
Instagram	3.00%	3.89%	<0.001	Yes	B
TikTok	2.45%	4.06%	<0.001	Yes	B
Google Search	8.88%	8.52%	0.2226	No	No significant difference
Email	11.76%	11.07%	0.0225	Yes	A

Reading the results by channel:

- Facebook, Instagram, TikTok: Variant B's conversion-rate lift is statistically significant, and it also lowers CAC and raises ROAS in every case — this is a clean, low-risk decision to fully roll out Variant B.
- Google Search: no significant difference between variants ($p = 0.22$) — Variant A numerically leads on CAC/ROAS but not by enough to justify retiring Variant B outright. Treat this as inconclusive, not as a win for A.
- Email: Variant A significantly outperforms Variant B ($p = 0.023$) on conversion rate — the one channel where the new creative underperforms the original. Both variants are so cheap that the dollar impact is small, but Variant A is the statistically sound default here.

Trends Over the Test Window

TikTok is scaling fast

Weekly TikTok impression volume rose from 97,636 in week 1 to 151,924 in week 13 — a 56% increase over the 13-week window — while CAC still improved (Variant B CAC of \$3.22 vs. Variant A's \$8.34). TikTok is the rare case

of a channel getting both bigger and more efficient at the same time, which makes it the strongest candidate for incremental budget.

Google Search creative is fatiguing

Average weekly CTR on Google Search declined from about 4.20% in the first three weeks of the test to about 3.68% in the final three weeks — a roughly 12% relative decline — consistent with ad fatigue as the same creative runs longer. This context matters when reading the "no significant difference" A/B result above: it may reflect a channel where both variants are aging, not a channel where the creative choice doesn't matter.

Email and Instagram were comparatively stable

Neither Email nor Instagram showed a directional trend over the window; their differences are driven by variant and channel economics rather than time.

Recommendations

1. Roll out Variant B on Facebook, Instagram, and TikTok

The lift is statistically significant on all three channels and improves CAC and ROAS simultaneously — this is the lowest-risk, highest-confidence action available from this test. Retire Variant A creative spend on these three channels.

2. Keep Variant A on Email

Variant A significantly outperforms Variant B on conversion rate for Email. Given Email's near-zero cost structure the dollar stakes are small, but there's no reason to run the underperforming variant once the data says otherwise.

3. Re-test Google Search creative rather than declaring a winner

The result is a statistical tie, and CTR decay over the window suggests both variants may simply be aging. Rather than picking a "winner" from an inconclusive test, refresh the creative on this channel and re-run the comparison — that will generate a cleaner read than continuing to run against a fatiguing baseline.

4. Shift incremental budget toward TikTok, away from Instagram

Instagram is the weakest channel on CAC (\$8.65) and ROAS (6.35x) even after adopting Variant B. TikTok is both cheaper per conversion (\$4.64 blended CAC) and growing in scale. As an illustrative example: reallocating \$3,000 from Instagram to TikTok, at each channel's blended CAC, trades roughly 347 Instagram conversions for roughly 647 TikTok conversions — a net gain of about 300 conversions and roughly \$16,000 in incremental revenue from the same budget, before accounting for any further TikTok scale effects.

This is a directional illustration, not a guarantee — reallocating spend can change a channel's marginal efficiency (diminishing returns at higher scale), so any shift should be tested incrementally and re-measured rather than moved all at once.

5. Lock the date format at the source

Standardize on ISO (YYYY-MM-DD) date exports from every ad platform/reporting tool feeding this pipeline. It removes the small but avoidable ambiguity currently affecting about 2% of rows and makes the cleaning script's date-parsing step unnecessary going forward.

Appendix: Files in This Analysis

- 01_generate_data.R — simulates the raw, messy campaign export
- 02_clean_and_analyze.R — cleans data, computes KPIs, runs significance tests
- campaign_daily_clean.csv — cleaned daily grain data
- ab_test_summary.csv — channel × variant totals
- ab_significance_results.csv — z-test results per channel
- channel_summary.csv — channel-level rollup
- Dashboard.twb — Tableau workbook (Daily Trend, Spend vs. Conversions, CAC by Channel, A/B Conversion Rate Comparison)